HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **MES recipes** for electronic batch records in a large-scale biotech manufacturing facility, translating business requirements into configurable system logic.
- Configured and supported **interfaces between MES and other systems** including ERP, automation control systems, and laboratory data platforms, ensuring seamless data flow.
- Investigated and resolved production issues within the MES environment, determining root causes and recommending solutions to enhance system stability and performance.
- Developed high-quality, well-documented **MES configurations** adhering to company Quality Standards and Good Manufacturing Practices (GMP), facilitating easier maintenance and future updates.
- Generated computer system validation documentation in accordance with site procedures and GMP guidelines, supporting regulatory compliance for pharmaceutical production.
- Collaborated with site teams, including quality control and business analysts, to gather requirements, perform testing, and ensure successful implementation of MES system changes.
- Containerized **Java Spring Boot services** using Docker and deployed them to an AWS EC2 instance managed by Jenkins for CI/CD, improving deployment frequency by 40%.

SKILLS

MES & Manufacturing – MES Recipe Authoring, Emerson Syncade, Electronic Batch Records, Pharma 4.0, Manufacturing Execution Systems

Programming & Scripting – Visual Basic, Python, Java, Spring Boot, SQL

Configuration & Deployment – Configuration Management, System Design, Custom Off The Shelf (COTS) Implementation, Jenkins, Docker

Databases – MS SQL Server, PostgreSQL, MySQL, Oracle

Quality & Compliance – Quality Standards, GMP Guidelines, Computer System Validation, Regulatory Compliance

Core Engineering & Support – IT Application Support, Requirements Gathering, System Integration Testing, Problem Investigation, Change Management

PROJECTS

MES Recipe Configuration for Pharmaceutical Production

- Developed and configured **MES recipes** for a simulated API manufacturing process, focusing on accurate material dispensing and batch record generation.
- Integrated the MES recipe configuration with a mock **ERP system** via simulated REST API calls to manage material master data and production orders.
- Authored detailed design documentation and test cases for the MES recipes, ensuring alignment with hypothetical **GMP guidelines** and validation requirements.

Java Spring Boot Service for Manufacturing Data Ingestion

- Built a **Java Spring Boot** application to ingest and process manufacturing data streams from simulated equipment sensors, achieving p95 latency below 200ms.
- Designed a **PostgreSQL database schema** to store time-series manufacturing data, supporting efficient querying for historical analysis and reporting.
- Containerized the Spring Boot application using **Docker** and deployed it to an AWS EC2 instance, establishing a basic CI/CD pipeline with Jenkins for automated builds and deployments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)